



Analyzing Developer Trends: Insights from the Stack Overflow Survey

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OUTLINE



- **Executive Summary**
- **Introduction**
- **Methodology**
- **Results**
 - Visualization – Charts
 - Dashboard
- **Discussion**
 - Findings & Implications
- **Conclusion**
- **Appendix**

EXECUTIVE SUMMARY



- **Data Description and Contextualization**
- **Description of methodology**
 - Data Gathering
 - Exploratory Analysis
 - Data Visualization
- **Presentation of the results are supported with graphs and trends**
- **Discussion of overall findings and implications regarding the results**
- **Final conclusion of the Analysis**

INTRODUCTION



- The project focuses on analyzing data from the Stack Overflow Developer 2019 Survey
- Identify trends in programming languages, tools, and technologies from the survey results of about 90,000 respondents
- The objective is to provide actionable insights for stakeholders, such as tech companies and educational institutions, to align their strategies with industry trends
- The findings presented in the form of dashboards helped us understand
 - The top 10 languages, databases, platforms, and web frameworks currently used
 - The anticipated learning trends in languages, databases, platforms, and web frameworks for the next year
 - Demographic characteristics of the overall respondents of survey

METHODOLOGY



- All the data extraction, cleaning, comparison and presentation were done using python and its appropriate libraries
- Data collection and exploration
 - APIs
 - Web Scrapping using BeautifulSoup
 - Request libraries
- Data Wrangling
- Exploratory Data Analysis(EDA)
 - Analyzing Data Distribution
 - Handling Outliers
 - Correlations
- Data Visualization
 - Data distribution, relationships, composition and comparison using plots, charts and graphs
 - Dashboards
 - 3 separate tabs under a single dashboard using IBM Cognos Analytics
 - Current Technology Usage
 - Future Technology Trend
 - Demographics

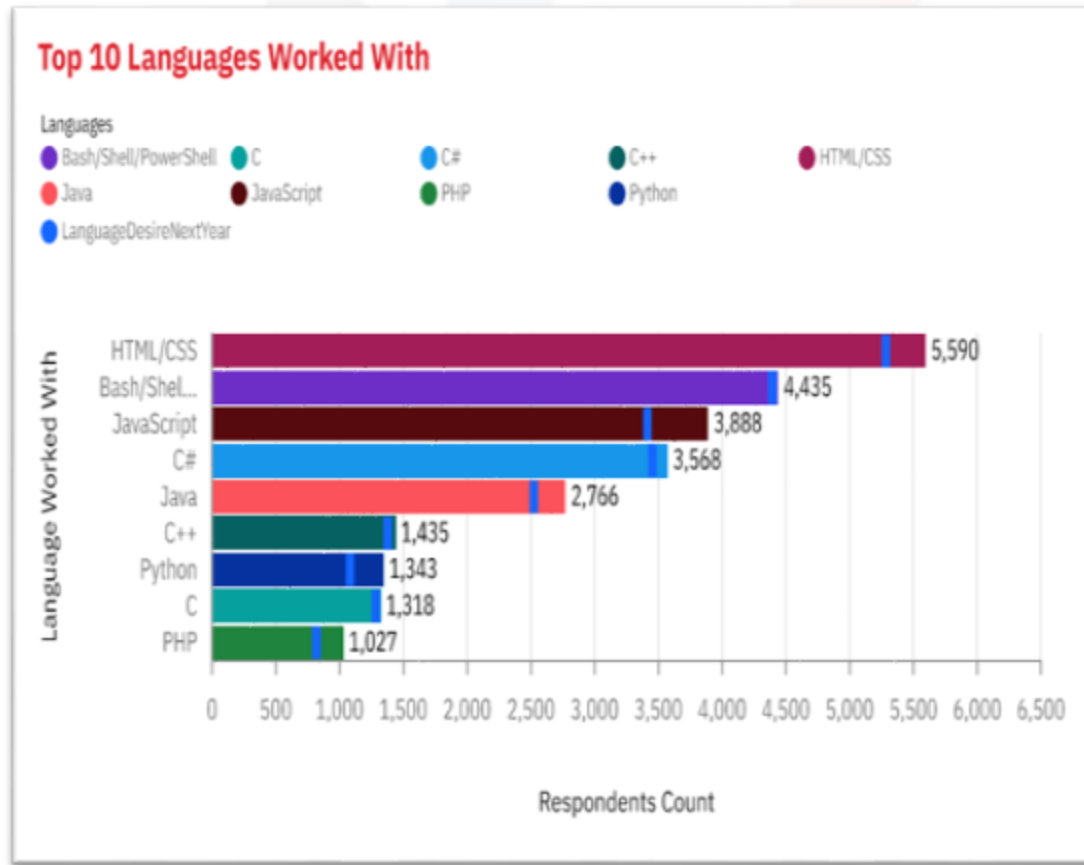
RESULTS



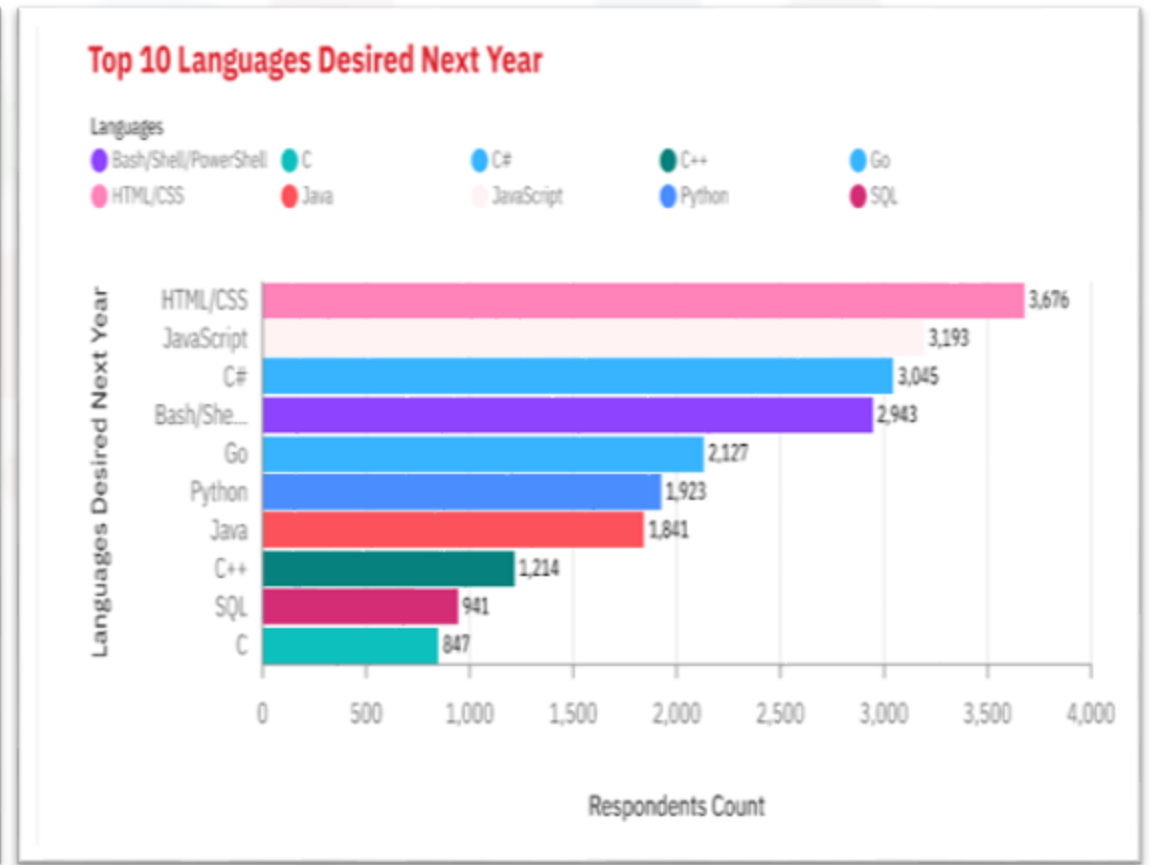
The analysis revealed that Python is the most popular programming language among developers, followed by JavaScript and HTML/CSS. The data also showed that developers highly value learning new technologies and that job satisfaction is generally high among respondents.

CURRENT TECHNOLOGY TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS

FINDINGS

- HTML | CSS keeps as leading language
- Python is the fastest growing language
- Growing interest for JavaScript

IMPLICATIONS

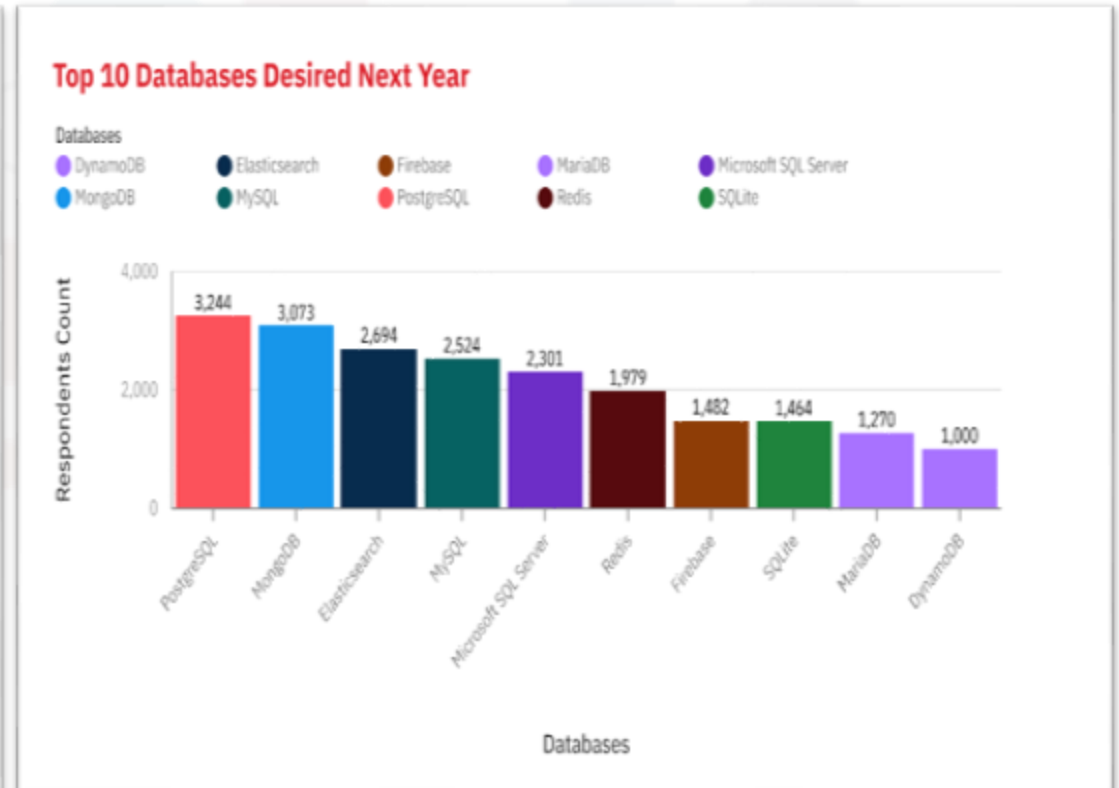
- Possible developers migration from Bash to JavaScript

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS

FINDINGS

- MySQL is the most prominently used database
- Limited interest in Firebase and DynamoDB
- Fast growing interest in Redis, PostgreSQL and MariaDB

IMPLICATIONS

- Firebase and DynamoDB are slightly losing ground in the market
- PostgreSQL is projected to gain significant ground in the market

DASHBOARD



The GitHub link of the Cognos dashboard goes here:

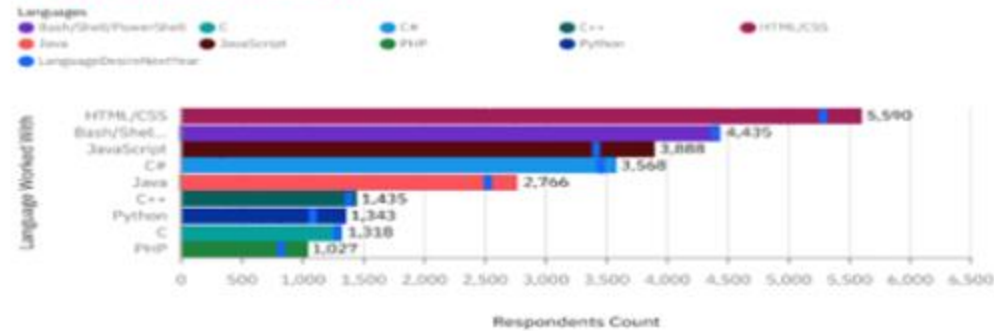
<https://github.com/WaqasHKhan/IBM-Data-Analyst-Capstone-Project/blob/cac0e3518b75a6c177aec9d140c5fe833d1d9ff3/IBM%20Cognos%20Analytics%20Dashboard.pdf>

CURRENT TECHNOLOGY TRENDS

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Current Technology Usage

Top 10 Languages Worked With

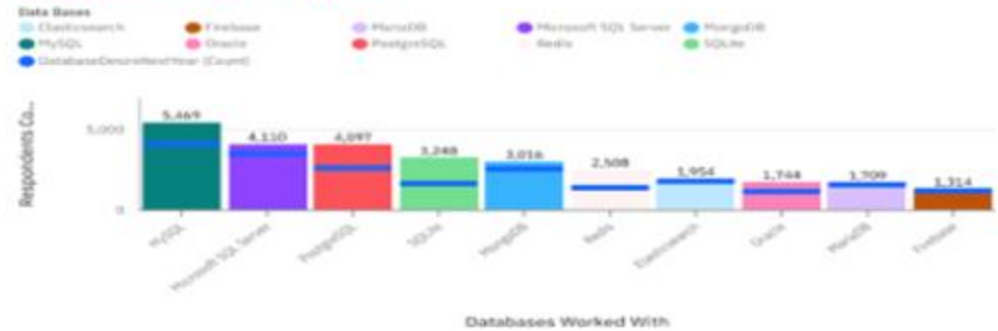


Platform Worked With



Final Dashboard

Top 10 Databases Worked With



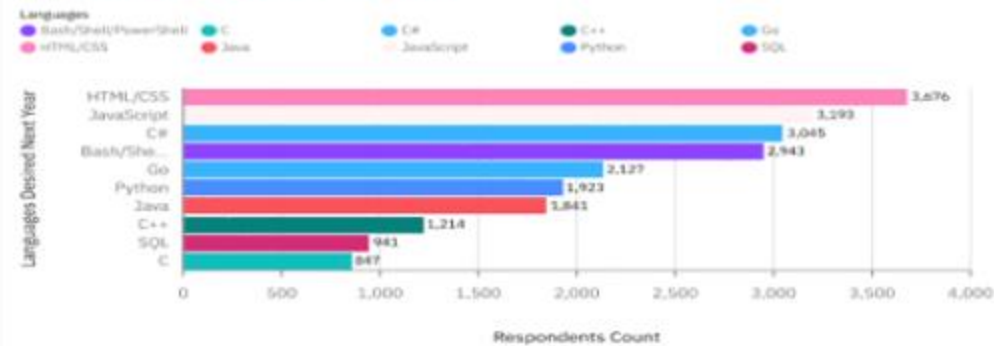
Top 10 Web Frame Worked With



FUTURE TECHNOLOGY TRENDS

Future Technology Trend

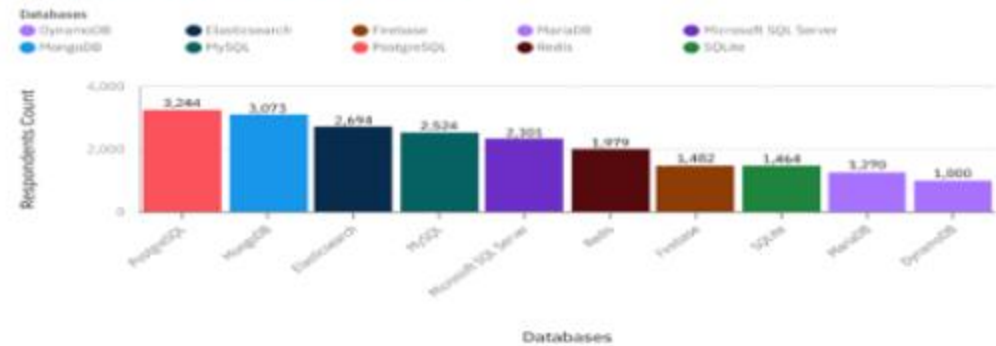
Top 10 Languages Desired Next Year



Platform Desire Next Year



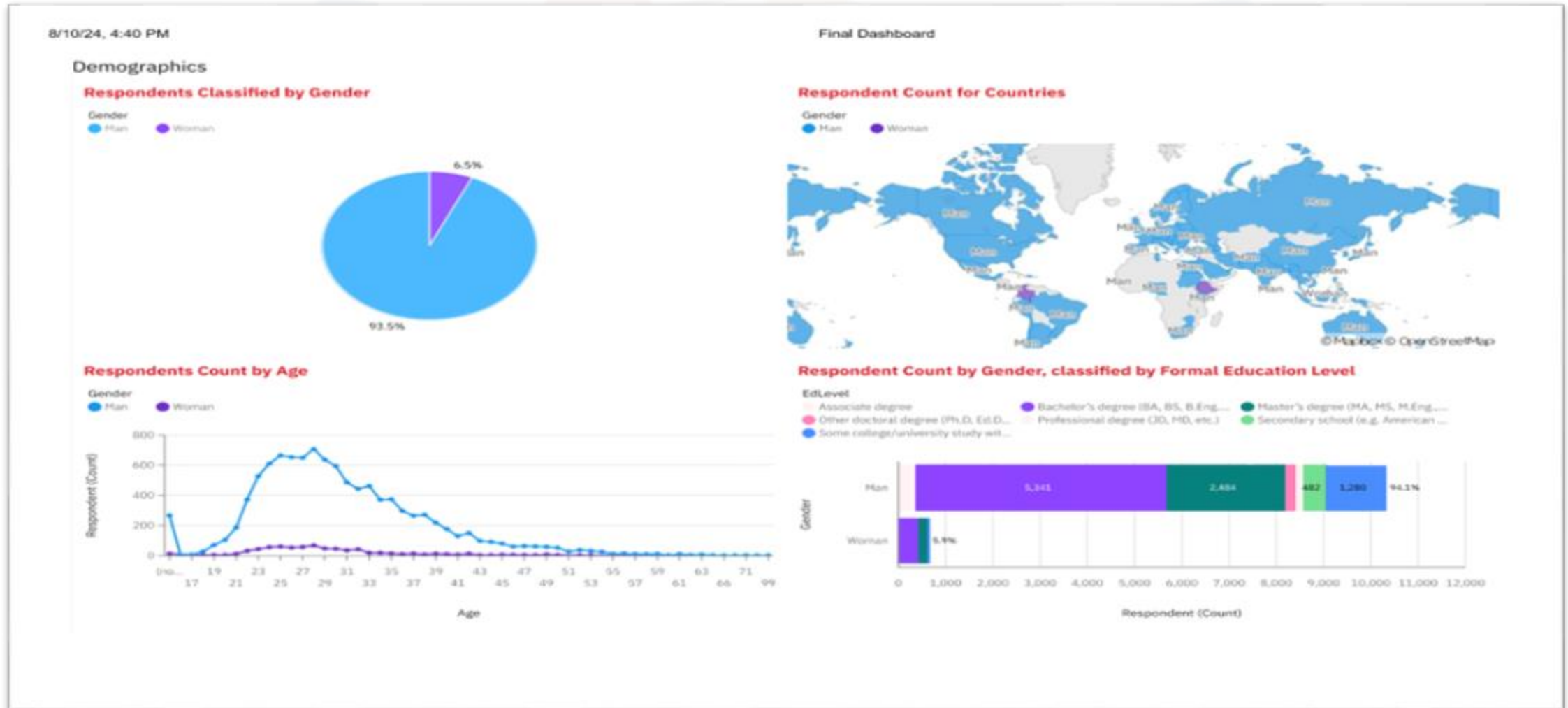
Top 10 Databases Desired Next Year



Top 10 Web Frame Desire Next Year



DEMOGRAPHICS



DISCUSSION



The results indicate a strong preference for Python among developers, which aligns with its growing use in data science and machine learning. The high job satisfaction suggests that the tech industry is providing fulfilling career opportunities. However, the data also highlights areas for improvement, such as gender diversity in tech.

OVERALL FINDINGS & IMPLICATIONS

FINDINGS

- JavaScript and MySQL most booming languages and databases among Developers
- Developers are predominantly young men at 90%
- A high majority of developers are found in developed countries

IMPLICATIONS

- JavaScript web frames gaining followers and PostgreSQL database is projected to be more in use in future
- Global polarization of developers location and gender
- Postgrad education level isn't much seems not to be a prerequisites in the Developers field as majority are undergraduates

CONCLUSION

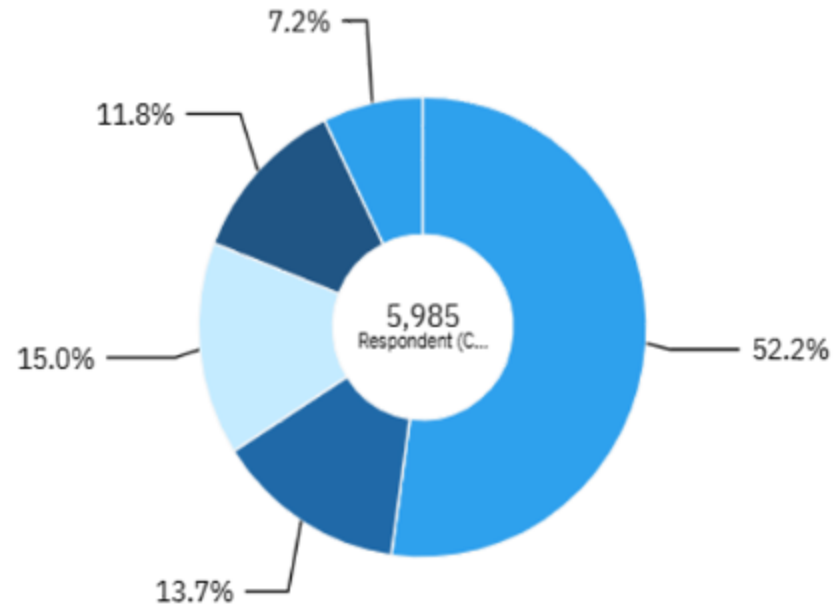


- It's crucial to stay updated and continuously upskill in the tech sector, as trends evolve over time.
- The insights suggest that stakeholders should make informed decisions regarding technology adoption and workforce development.
- Future research could investigate additional factors that influence developer satisfaction and career choices.
- Increasing the participation of women in STEM, particularly in tech, is essential to close the significant gender gap in this field.

APPENDIX



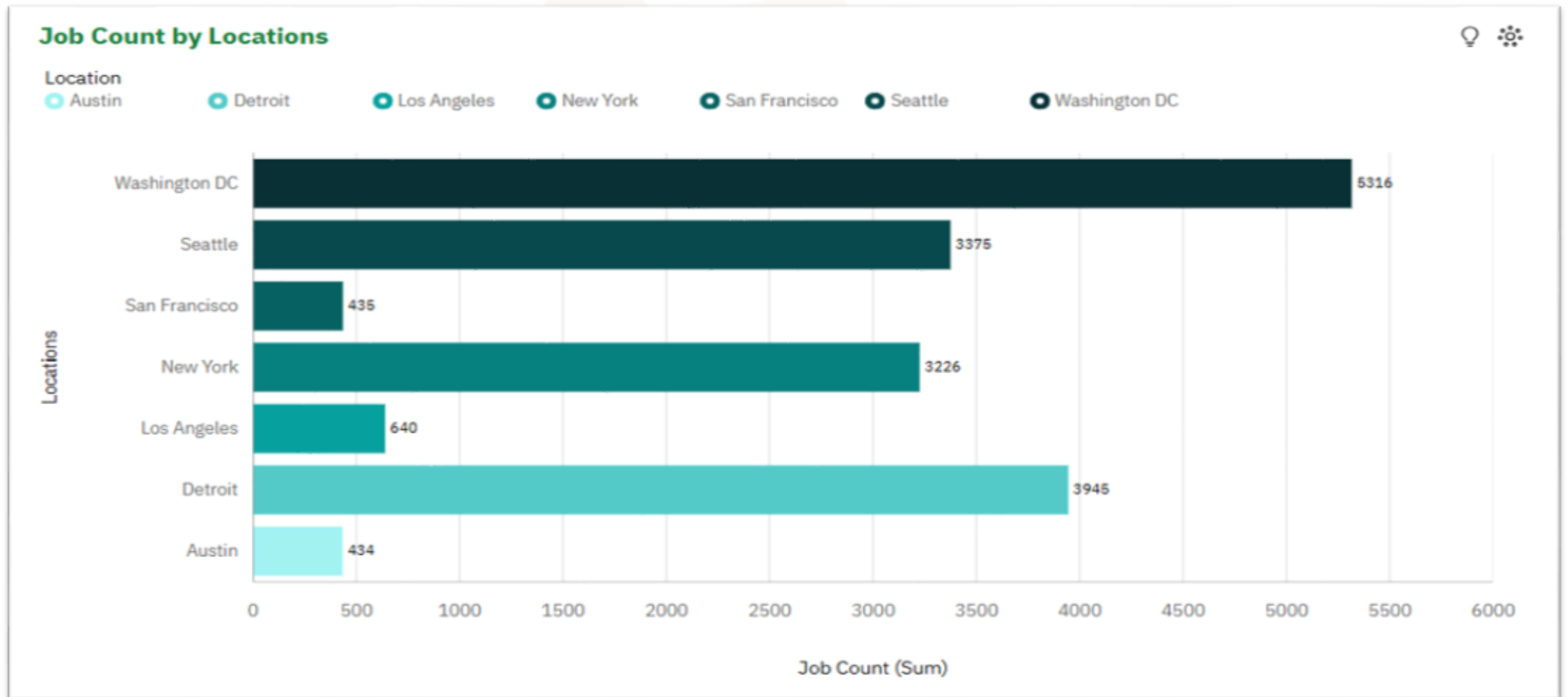
Respondents Count by Countries



Country

United States	3,127
United Kingdom	821
India	897
Germany	708
Canada	432

JOB POSTINGS



POPULAR LANGUAGES

